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Elastic member and elastic device

Abstract

An elastic member having a plate-like shape, includes a plurality of disc springs arranged side by side in a plane; and a support part integrally provided with the plurality of disc springs in the plane and around the plurality of disc springs, and supporting the plurality of disc springs.

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Background/Summary

TECHNICAL FIELD

(1) The present invention relates to an elastic member and an elastic device.

BACKGROUND ART

(2) Patent Document 1 discloses a technology for fixing a semiconductor module to a heat sink in a state where the heat radiation surface of the semiconductor module is pressed against the heat sink, by sandwiching the semiconductor module between the heat sink and a spring-like pressing plate, in a semiconductor device.

CITATION LIST

Patent Document

(3) Patent document 1: Japanese Unexamined Patent Application Publication No. 2013-110241

SUMMARY OF INVENTION

Technical Problem

(4) However, because the technology of Patent Document 1 uses a simple thin metal plate as a pressing plate, the semiconductor module cannot be pushed strongly against the heat sink.

Therefore, the technology of Patent Document 1 may cause rattling of the semiconductor module or decrease the cooling efficiency of the semiconductor module.

(5) In recent years, in order to realize miniaturization of various devices such as semiconductor modules, an elastic member which can be arranged in a narrow gap and which can apply a high load is required; however, with the pressing plate of Patent Document 1, even if the pressing plate can be arranged in a narrow gap, the pressing plate cannot apply a high load.

(6) In order to solve the above-mentioned problem, an object of the present invention is to provide an elastic member capable of applying a high load in a narrow gap.

Solution to Problem

(7) In order to solve the problems described above, the elastic member according to one embodiment is an elastic member having a plate-like shape, including a plurality of disc springs arranged side by side in a plane; and a support part integrally provided with the plurality of disc springs in the plane and around the plurality of disc springs, and supporting the plurality of disc springs.

Advantageous Effects of Invention

(8) According to an elastic member according to an embodiment, it is possible to provide an elastic member capable of applying a high load in a narrow gap.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1 is an exploded perspective view of an elastic device according to an embodiment.

(2) FIG. 2 is a partial enlarged side view of an elastic device according to an embodiment.

(3) FIG. 3 is an exploded perspective view illustrating another configuration example of an elastic device according to an embodiment.

(4) FIG. 4 is a partial enlarged side view illustrating another configuration example of an elastic device according to an embodiment.

(5) FIG. 5 is a table illustrating an analysis result of load characteristics of an elastic device according to an embodiment.

(6) FIG. 6 is a graph illustrating an analysis result of load characteristics of an elastic device according to an embodiment.

(7) FIG. 7 is a diagram illustrating an example of use of an elastic member according to an

embodiment.

(8) FIG. 8 is a diagram illustrating a modified example of an elastic member according to an embodiment.

(9) FIG. 9 is a diagram illustrating a modified example of an elastic device according to an embodiment.

DESCRIPTION OF EMBODIMENTS

(10) An embodiment of the present invention will now be described with reference to the drawings. Configuration of Elastic Device 10

(11) FIG. 1 is an exploded perspective view of an elastic device 10 according to an embodiment.

FIG. 2 is a partial enlarged side view of the elastic device 10 according to an embodiment.

(12) As illustrated in FIG. 1, the elastic device 10 according to an embodiment is generally thin and planar. The elastic device 10 includes two elastic members 100 (elastic members 100-1, 100-2) and an intermediate plate 20. The elastic device 10 has a laminated structure in which the intermediate plate 20 is sandwiched between the two elastic members 100. Although the two elastic members 100 are parts identical to each other, they are used in a state in which the front and back of each other are inverted.

(13) In the present embodiment, for convenience, the thickness direction of the elastic device 10 is set as the vertical direction (Z-axis direction), the longitudinal direction of the elastic device 10 is set as the horizontal direction (X-axis direction), and the transverse direction of the elastic device 10 is set as the front back direction (Y-axis direction).

(14) The elastic member 100 is a metal flat plate member having a rectangular outer shape with the longitudinal direction in the horizontal direction (X-axis direction) in a plan view. The elastic member 100 has a plurality of disc springs 110 and a support part 120.

(15) The plurality of disc springs 110 are arranged in a line in the horizontal direction (X-axis direction) and in the same plane, in the frame of a base part 121 of the support part 120.

(16) The disc springs 110 are disc-shaped members having a circular opening 111 in the center and having elasticity in the vertical direction (Z-axis direction). The disc springs 110 have a shape that gradually rises in the vertical direction (Z-axis direction) from the outer periphery toward the inner periphery. That is, the disc spring 110 has a trapezoidal shape with a central top. Thus, the disc spring 110 is elastically deformable in the vertical direction (Z-axis direction) so that the vertical width is reduced by pressing the top (that is, the central portion where the opening 111 is provided). Further, the disc spring 110 can apply a high load to an object against which the disc spring 110 is pressed by its own elastic force.

(17) The number and arrangement of the disc springs 110 are not limited to those illustrated in FIG. 1. For example, in the example illustrated in FIG. 1, the elastic member 100 has six disc springs 110. However, the number of disc springs is not limited thereto, and the elastic member 100 may have five or fewer disc springs, or seven or more disc springs 110. In the example illustrated in FIG. 1, the elastic member 100 has a plurality of disc springs 110 arranged in a row in the longitudinal direction (X-axis direction); however, the arrangement is not limited thereto, and the elastic member 100 may have a plurality of disc springs 110 arranged in plural rows in the longitudinal direction (X-axis direction).

(18) In the example illustrated in FIG. 1, the elastic member 100 has a plurality of disc springs 110 arranged in a straight line, but the arrangement is not limited thereto, and the elastic member 100 may have two adjacent disc springs 110 offset in the transverse direction (Y-axis direction). In this case, the elastic member 100 can arrange the same number of disc springs 110 within a shorter range in the longitudinal direction (X-axis direction).

(19) The support part 120 is shaped as a flat-plate rectangular frame. The support part 120 is integrally provided with the plurality of disc springs 110, in the same plane as the plurality of disc springs 110 and around the plurality of disc springs 110, and supports the plurality of disc springs 110. The support part 120 has a frame-like base part 121 and supports the plurality of disc springs

110 within the frame of the base part **121**. Specifically, the support part **120** has a connection part **122** connecting the base part **121** and the disc spring **110**, with respect to each of the plurality of disc springs **110** within the frame of the base part **121**. In the example illustrated in FIG. 1, the support part **120** has connection parts **122** connecting disc springs **110** to the portion of the base part **121** on the positive Y-axis side and connection parts **122** connecting disc springs **110** to the portion of the base part **121** on the negative Y-axis side, with these connection parts **122** being provided alternately in the horizontal direction (X-axis direction).

(20) In the example illustrated in FIG. 1, the base part **121** has a rectangular frame shape with the horizontal direction (X-axis direction) as the longitudinal direction. However, the base part **121** is not limited thereto, and the base part **121** may have a frame shape other than a rectangular shape (for example, square, circular, etc.). The base part **121** may have a shape other than a frame shape (for example, linear).

(21) The elastic member **100** is formed by using any metal material (for example, stainless steel). By press-processing one metal plate, the elastic member **100** can be formed at once to have a plurality of disc springs **110** and the support part **120**. For example, in the present embodiment, the elastic member **100** is formed at once to have a plurality of disc springs **110** and the support part **120** by press-processing a stainless steel plate having a thickness of 0.5 [mm], for example. It is also possible to manufacture a plurality of elastic members **100** at once by press-processing a single metal plate.

(22) The intermediate plate **20** is a plate-like member having a rectangular shape whose longitudinal direction is in the horizontal direction (X-axis direction) in a plan view. The intermediate plate **20** is formed by using any metal material (for example, stainless steel).

(23) As illustrated in FIGS. 1 and 2, in the elastic device **10**, the elastic member **100-1** provided on the upper side (positive Z-axis side) is arranged so that the tops of all the disc springs **110** face downward. On the other hand, in the elastic device **10**, the elastic member **100-2** provided on the lower side (negative Z-axis side) is arranged so that the tops of all the disc springs **110** face upward.

(24) In the elastic device **10**, as illustrated in FIG. 2, the intermediate plate **20** is arranged between the two elastic members **100** so as to be sandwiched between the tops of the disc springs **110** facing downward and provided on the upper side (positive Z-axis side) and the tops of the disc springs **110** facing upward and provided on the lower side (negative z-axis side).

(25) The elastic member **100** according to one embodiment has six disc springs **110**, and, therefore, it is possible to obtain a load six times larger than that obtained by one disc spring **110**, while being thin overall.

(26) Further, the elastic member **100** according to one embodiment is thin and flat overall, and, therefore, it is possible to be installed within a narrow gap and to apply a high load evenly over a wide range within the gap.

(27) Further, the elastic member **100** according to one embodiment has a plurality of disc springs **110**, and, therefore, the overall load can be easily adjusted by adjusting the number, the arrangement, the shape, the outer diameter, the inner diameter, etc. of the disc springs **110**. In the example illustrated in FIG. 1, the plurality of disc springs **110** have the same shape and the same size, but the plurality of disc springs **110** are not limited thereto, and the plurality of disc springs **110** may have different shapes and different sizes.

(28) In the elastic member **100** according to one embodiment, the plurality of disc springs **110** are supported by the support part **120**, and, therefore, the plurality of disc springs **110** can be maintained at a predetermined position, and because the plurality of disc springs **110** will not be separated from each other, the handling of the plurality of disc springs **110** can be facilitated.

(29) In the elastic device **10** according to one embodiment, the two elastic members **100** are overlapping, and, therefore, the overall deflection amount can be doubled while the height dimension of each disc spring **110** is reduced. In the elastic device **10** according to one

embodiment, by reducing the height dimension of each disc spring **110**, it is possible to make it difficult to invert the disc spring **110**.

(30) Further, in the elastic device **10** according to one embodiment, two elastic members **100** are overlapped so that the tops of the plurality of disc springs **110** of the elastic member **100-1** on the upper side (positive z-axis side) and the tops of the plurality of disc springs **110** of the elastic member **100-2** on the lower side (positive Z-axis side) face each other. Therefore, the elastic device **10** according to one embodiment can double the overall deflection amount while making the overall load obtained equal to the load obtained by one elastic member **100**.

(31) The elastic device **10** according to one embodiment has a thin and flat shape as a whole, and, therefore, the elastic device **10** can be installed in a narrow gap, and a high load can be applied evenly over a wide range in the gap.

(32) Further, the elastic device **10** according to one embodiment has an intermediate plate **20** between two elastic members **100**, and, therefore, the tops of the respective disc springs **110** can be pressed against the surface of the intermediate plate **20** reliably and evenly, that is, the load from the respective disc springs **110** can be received reliably by the intermediate plate **20**.

(33) The elastic device **10** according to one embodiment has an intermediate plate **20** between the two elastic members **100**, and, therefore, the overall thickness can be easily adjusted by adjusting the thickness of the intermediate plate **20**.

(34) In the elastic device **10** according to one embodiment, the two elastic members **100** may be overlapped so that the bottom portions of the plurality of disc springs **110** of the elastic member **100-1** on the upper side (positive Z-axis side) and the bottom portions of the plurality of disc springs **110** of the elastic member **100-2** on the lower side (positive Z-axis side) face each other. In this case, the elastic device **10** according to one embodiment can double the overall deflection amount while making the overall load obtained equal to the load obtained by one elastic member **100**.

(35) Further, the elastic device **10** according to one embodiment may be provided with two elastic members **100** overlapping, so that the bottom of each of the plurality of disc springs **110** of the elastic member **100-1** on the upper side (positive Z-axis side) and the top of each of the plurality of disc springs **110** of the elastic member **100-2** on the lower side (positive Z-axis side) face each other. In this case, the elastic device **10** according to one embodiment can double the overall deflection amount while making the overall load obtained to be double that obtained by one elastic member **100**.

(36) On the other hand, the elastic device **10** according to one embodiment may be provided with two elastic members **100** overlapping so that the tops of the plurality of disc springs **110** of the elastic member **100-1** on the upper side (positive Z-axis side) and the bottoms of the plurality of disc springs **110** of the elastic member **100-2** on the lower side (positive Z-axis side) face each other. In this case, the elastic device **10** according to one embodiment can double the overall deflection amount while making the overall load obtained be double that obtained by the single elastic member **100**.

Another Configuration Example of the Elastic Device **10**

(37) FIG. **3** is an exploded perspective view illustrating another configuration example of the elastic device **10** according to an embodiment. FIG. **4** is a partial enlarged side view illustrating another configuration example of the elastic device **10** according to an embodiment.

(38) The elastic device **10** illustrated in FIGS. **3** and **4** has a flat plate-like rigid body **30** at the top (upper side of the elastic member **100-1**) and at the bottom (lower side of the elastic member **100-2**). That is, the elastic device **10** illustrated in FIG. **4** has a configuration in which the elastic device **10** illustrated in FIG. **1** is sandwiched between two rigid bodies **30** (rigid bodies **30-1**, **30-2**).

(39) Each of the two rigid bodies **30** has a first surface **31** facing inward and facing the plurality of disc springs **110**, and a second surface **32** facing outward and facing an object to receive a load applied from the elastic device **10**, on the reverse side of the first surface. Each of the two rigid

bodies **30** is formed with a constant thickness so as to be difficult to elastically deform, using any metal material.

(40) The elastic device **10** illustrated in FIG. 4, having two rigid bodies **30**, can receive the load applied from the plurality of disc springs **110** on the entire first surface **31** of each of the two rigid bodies **30**, and apply the load evenly to the object by the entire second surface of each of the two rigid bodies **30**.

Load Characteristics of the Elastic Device **10**

(41) FIG. 5 is a table indicating an analysis result of load characteristics of the elastic device **10** according to an embodiment. FIG. 6 is a graph indicating an analysis result of load characteristics of the elastic device **10** according to an embodiment.

(42) The length of the elastic device **10** of the present analysis result in the longitudinal direction (X-axis direction) is 135 [mm], and the width in the transverse direction (Y-axis direction) is 35 [mm].

(43) The thickness of the elastic device **10** of the present analysis result is 2.72 [mm] when not pressed, which is extremely thin.

(44) In the elastic device **10** of the present analysis result, the height, the outer diameter, and the inner diameter of each disc spring **110** are adjusted so as to satisfy the following performance requirements (1) and (2). A load 930 [N] can be obtained with a displacement amount of 0.3 [mm] .

. . (1) A load 1300 [N] can be obtained with a displacement amount of 0.82 [mm] . . . (2)

(45) As illustrated in FIGS. 5 and 6, it was confirmed that the elastic device **10** of the present analysis result can obtain an extremely high load of 930 [N] when the thickness at the time of pressing is 2.42 [mm] (that is, the displacement amount is 0.3 [mm]), and can satisfy the required performance.

(46) As illustrated in FIGS. 5 and 6, it was confirmed that the elastic device **10** of the present analysis result can obtain an extremely high load of 1300 [N] when the thickness at the time of pressing is 1.9 [mm] (that is, the displacement amount is 0.82 [mm]), and can satisfy the required performance.

(47) As illustrated in FIGS. 5 and 6, the elastic device **10** of the present analysis result has a first load characteristic from a displacement amount of 0 to 0.3 [mm] (an example of the “first value”) and a second load characteristic from a displacement amount of 0.3 [mm] to 0.82 [mm] (an example of the “second value”), and the first load characteristic has a higher rate of increase in load than the second load characteristic. That is, the elastic device **10** of the present analysis result can obtain a high load with a relatively small displacement amount after the pressing is started.

One Use Example of the Elastic Member **100**

(48) FIG. 7 is a diagram illustrating one use example of the elastic member **100** according to an embodiment. As illustrated in FIG. 7, in the elastic member **100** according to an embodiment, a plurality of disc springs **110** are connected by a support part **120**, and, therefore, the three disc springs **110** and the other three disc springs **110** can be overlapped by folding back the support part **120** at an intermediate position in the longitudinal direction (X-axis direction), and the overall thickness can be doubled.

One Modified Example of the Elastic Member **100**

(49) FIG. 8 is a diagram illustrating an elastic member **100A** which is one modified example of the elastic member **100** according to an embodiment.

(50) In a plan view, the elastic member **100A** is a metal flat plate member whose longitudinal direction is in the horizontal direction (X-axis direction). The elastic member **100A** has a first disc spring row **L1**, a second disc spring row **L2**, and a support part **120A**.

(51) The first disc spring row **L1** and the second disc spring row **L2** are provided in parallel, with the support part **120A** between them. The first disc spring row **L1** is provided on the positive Y-axis side of the support part **120A**. The second disc spring row **L2** is provided on the negative Y-axis side of the support part **120A**.

(52) Each of the first disc spring row L1 and the second disc spring row L2 has a plurality of disc springs **110** arranged in a row in a horizontal direction (X-axis direction) in the same plane. The disc spring **110** provided in the elastic member **100A** is the same as the disc spring **110** provided in the elastic member **100**.

(53) In the first disc spring row L1 and the second disc spring row L2, the number of disc springs **110** is not limited to those illustrated in FIG. 8. For example, in the example illustrated in FIG. 8, the first disc spring row L1 and the second disc spring row L2 each have five disc springs **110**. However, the number of disc springs is not limited thereto, and the first disc spring row L1 and the second disc spring row L2 may each have four or less or six or more disc springs **110**.

(54) In the example illustrated in FIG. 8, the first disc spring row L1 and the second disc spring row L2 may each have a plurality of disc springs **110** arranged in a straight line, but the arrangement is not limited thereto, and the first disc spring row L1 and the second disc spring row L2 may each have two adjacent disc springs **110** offset in the transverse direction (Y-axis direction). In this case, the same number of disc springs **110** can be arranged in the first disc spring row L1 and the second disc spring row L2 within a shorter range in the longitudinal direction (X-axis direction).

(55) The support part **120A** is provided integrally with the plurality of disc springs **110** in the same plane as the plurality of disc springs **110** and between the first disc spring row L1 and the second disc spring row L2 to support the plurality of disc springs **110**. The support part **120A** has a base part **121** extending linearly in the horizontal direction (X-axis direction) and supports the plurality of disc springs **110** on both outer sides of the base part **121**. Specifically, the support part **120A** has a connection part **122** connecting the base part **121** and the disc spring **110**, for each of the plurality of disc springs **110** on both outer sides of the base part **121**. In the example illustrated in FIG. 8, the support part **120A** has a plurality of connection parts **122** connecting the plurality of disc springs **110** held by the first disc spring row L1 to the base part **121** on the first disc spring row L1 side (positive Y-axis side) relative to the base part **121**, and has a plurality of connection parts **122** connecting the plurality of disc springs **110** held by the second disc spring row L2 to the base part **121** on the second disc spring row L2 side (negative Y-axis side) relative to the base part **121**.

(56) The elastic member **100A** is formed by using any metal material (for example, stainless steel). By press-processing one metal plate, the elastic member **100A** can be formed at once to have a plurality of disc springs **110** and the support part **120A**. For example, in the present embodiment, the elastic member **100A** is formed at once to have a plurality of disc springs **110** and the support part **120A** by, for example, press-processing a stainless steel plate having a thickness of 0.5 [mm]. It is also possible to manufacture a plurality of elastic members **100A** at once by press-processing a single metal plate.

(57) The elastic device **10** according to one embodiment may use the elastic member **100A** instead of the elastic member **100**. Also in this case, the elastic device **10** according to one embodiment can achieve the same effect as when the elastic member **100** is used.

One Modified Example of the Elastic Device **10**

(58) FIG. 9 is a diagram illustrating one modified example of the elastic device **10** according to an embodiment. In the elastic device **10** according to an embodiment, two elastic members **100**, which are parts identical to each other, may be overlapping without inverting each other.

(59) For example, as illustrated in FIG. 9, in the elastic device **10** according to an embodiment, two elastic members **100** may be overlapping with an intermediate plate **20** therebetween so that the tops of the plurality of disc springs **110** of the elastic member **100-1** on the upper side (positive Z-axis side) face upward and the tops of the plurality of disc springs **110** of the elastic member **100-2** on the lower side (negative Z-axis side) face upward.

(60) In this case, the elastic device **10** according to one embodiment can double the overall deflection amount while making the overall load obtained equal to the load obtained by one elastic member **100**.

(61) Although the preferred embodiments of the present invention have been described in detail above, the present invention is not limited to these embodiments and can be modified or changed in various ways within the scope of the gist of the present invention described in the claims.

(62) For example, the elastic member **100**, **100A** may be used separately or incorporated into the elastic device **10**.

(63) The elastic device **10** may have a configuration in which three or more elastic members **100**, **100A** are overlapping. In this case, the elastic device **10** may have the intermediate plate **20** between two elastic members **100**, **100A** overlapping each other.

(64) The present international application claims priority to Japanese Patent Application No. 2022-089903 filed on Jun. 1, 2022, the entire contents of which are incorporated herein by reference.

REFERENCE SIGNS LIST

(65) **10** elastic device **20** intermediate plate **30**, **30-1**, **30-2** rigid body **100**, **100-1**, **100-2**, **100A** elastic member **110** disc spring **111** opening **120**, **120A** support part **121** base part **122** connecting part

Claims

1. An elastic member having a plate-like shape, comprising: a plurality of disc springs arranged side by side in a plane; and a support part integrally provided with the plurality of disc springs in the plane and around the plurality of disc springs, and supporting the plurality of disc springs, wherein the support part includes a base part having a frame-like shape and holds the plurality of disc springs within a frame of the base part such that each disc spring is connected at one end to the base part but unsupported at other ends by the support part without being connected to the base part.

2. The elastic member according to claim 1, wherein the base part has a rectangular frame-like shape.

3. The elastic member according to claim 2, wherein the plurality of disc springs are arranged side by side in the frame of the base part of the support part along a longitudinal direction of the base part.

4. An elastic member having a plate-like shape, comprising: a plurality of disc springs arranged side by side in a plane; and a support part integrally provided with the plurality of disc springs in the plane and around the plurality of disc springs, and supporting the plurality of disc springs, wherein the support part includes a base part having a linear shape, and holds the plurality of disc springs at both outer sides of the base part such that each disc spring is connected at one end to the base part but unsupported at other ends by the support part without being connected to the base part.

5. The elastic member according to claim 4, wherein the plurality of disc springs are arranged side by side along a longitudinal direction of the base part at both outer sides of the base part of the support part.

6. An elastic device comprising: two of the elastic members according to claim 5, the elastic members being provided to overlap each other.

7. The elastic device according to claim 6, wherein the two elastic members are provided to overlap each other such that tops of the plurality of disc springs provided in one elastic member of the two elastic members and tops of the plurality of disc springs provided in another elastic member of the two elastic members face each other.

8. The elastic device according to claim 7, further comprising: an intermediate plate arranged between the two elastic members.

9. The elastic device according to claim 6, wherein the two elastic members are provided to overlap each other such that tops of the plurality of disc springs provided in one elastic member of the two elastic members and tops of the plurality of disc springs provided in another elastic member of the

two elastic members are oriented in a same direction.

10. The elastic device according to claim 9, further comprising: an intermediate plate arranged between the two elastic members.

11. The elastic device according to claim 6, wherein: a first load characteristic represents a rate of load increase relating to a displacement amount of the elastic device when a thickness value of the elastic device at the time of being pressed is between 0 to a first value; and a second load characteristic represents the rate of load increase relating to a displacement amount of the elastic device when the thickness value of the elastic device at the time of being pressed is between the first value to a second value, wherein the first load characteristic has a higher rate of load increase than the second load characteristic.
